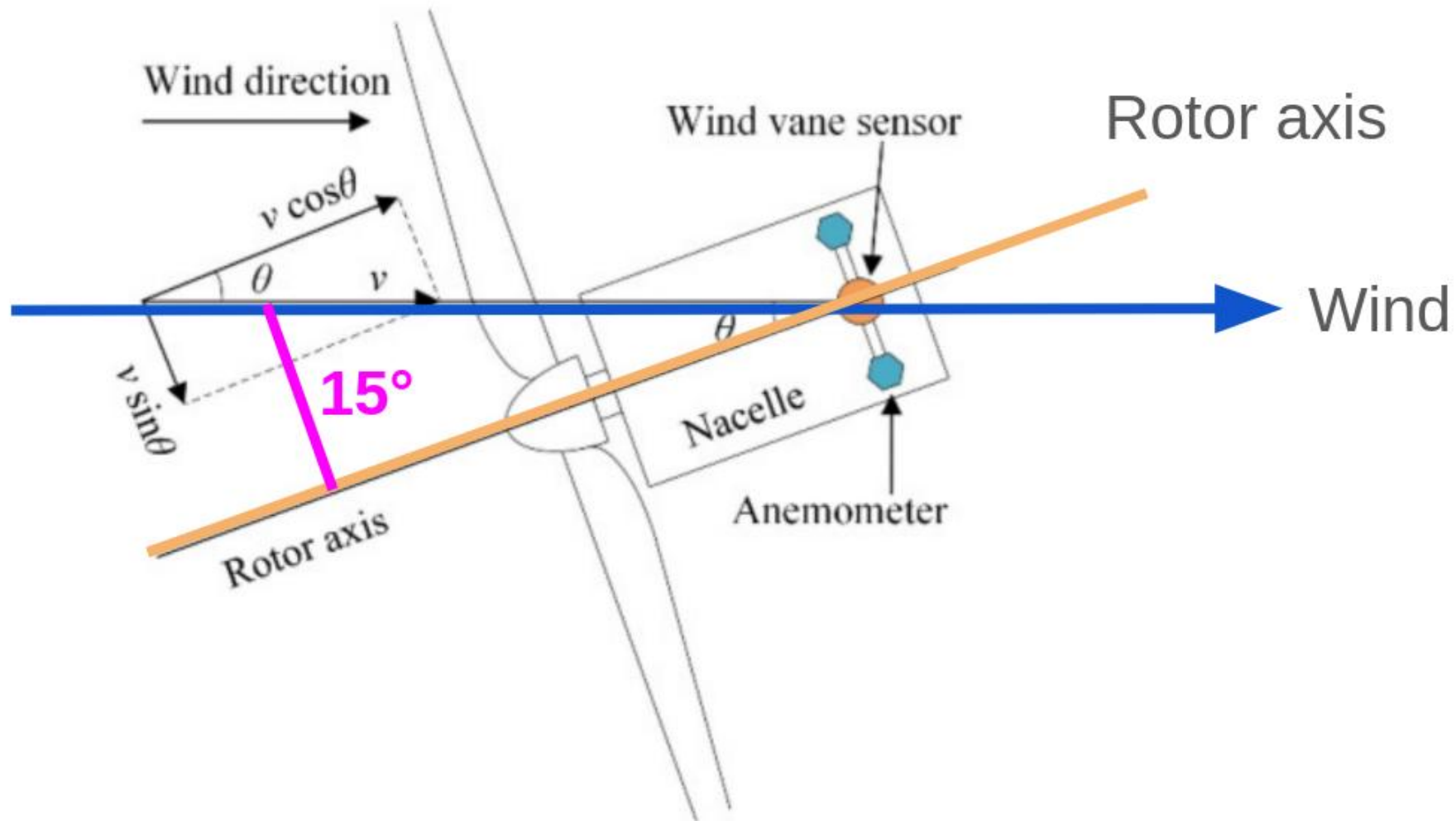


WeDoWind

How to determine the yaw-misalignment in a cost effective manner for NADARA

Energy Data Hackdays • 10.09.2026





A 15° misalignment can cost $\approx 10\%$ of power

What happens?

Wind must be aligned with the rotor to maximize energy production.

Today's approach

Add an extra sensor to measure yaw alignment.

Why not?

$\approx$ CHF 200k per turbine $\rightarrow \approx$ CHF 20M for 100 turbines.

So we asked: can we infer the missing signal from SCADA data we already have?

Use the fleet to identify persistent yaw states



Key idea: neighbours reveal state changes; the physics prior sets absolute level.

The data contains a usable signal

T0 • NO LABELS

PPP_WTG17 • held-out

blind target

731 validation dates

T3 • 3 TURBINES

PPP_WTG12 / 13 / 14

strict LOTO

PUBLIC VALIDATION • PPP_WTG17

1.44 **1.27**

RMSE (°) MAE (°)
ARI 1.00 • SHAPE 0.99° • BIAS -1.05°

T3 DEVELOPMENT LOTO

0.524 **0.358**

RMSE (°) MAE (°)

Result

Persistent yaw states can be inferred from existing SCADA—no additional sensor required.

Why it matters

A scalable, label-free workflow for wind farms.

THE TAKEAWAY

**Don't add hardware
before asking what the data already knows.**

WeDoWind • From existing SCADA data → hidden yaw insight

Thank you